

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide $K_4Fe(CN)_6$. After filtering, the precipitate is dried and weighed. The reactions which occur are $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$ $2ZnCl_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KCl$ If a 2 gram sample of rock yields 0.969 gram of $Zn_2Fe(CN)_6$, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight $Zn_2Fe(CN)_6 = 342.6$.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

</tool_call> reasoning process here

We are given :

- A 2 gram sample of rock yields 0.969 g of $Zn_2Fe(CN)_6$.
- We need to find the percentage of Zn in the original rock sample.

Step 1 : Use the molecular weights to find how much Zn is in the precipitate of $Zn_2Fe(CN)_6$.

Molecular weight of $Zn_2Fe(CN)_6 = 342.6$ g/mol

Atomic weight of Zn = 65.4 g/mol

So , 2 moles of Zn in $Zn_2Fe(CN)_6$ weigh :

$$2 \times 65.4 = 130.8 \text{ g of Zn}$$

So , in 342.6 g of $Zn_2Fe(CN)_6$ $\rightarrow$ 130.8 g of Zn

Thus , the fraction of Zn in $Zn_2Fe(CN)_6 = 130.8 / 342.6$

Now , the mass of Zn in 0.969 g of $Zn_2Fe(CN)_6$ is :

$$\text{Zn mass} = (130.8 / 342.6) \times 0.969$$

Calculate that :

$$\text{First , } 130.8 / 342.6 \approx 0.3817$$

$$\text{Then , } 0.3817 \times 0.969 \approx 0.3705 \text{ g of Zn}$$

This is the amount of Zn in the original 2 g rock sample .

Now , percentage of Zn in the rock = (mass of Zn / mass of sample) $\times 100$

$$= (0.3705 / 2) \times 100$$

$$= 18.525 \%$$

Closest value among the choices is :

F. 18 % Zn

< answer > 18 </ answer >